SIMATS ENGINEERING
ASSESSMENT TOOL 3: Application Based Questions

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED
CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2) Total Marks: 25
Question Count: 5

Code of Conduct
I S.SHAMAH ALI FATHIMA (192573002) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: 

Assessment Tasks

Application Based Questions

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.

2. An organization wants to store large volumes of backup data securely in the cloud.
   Discuss how cloud storage solutions ensure durability and data redundancy.
   How can encryption and access control protect sensitive backup data?
   Explain the cost advantages compared to traditional storage systems.
   Provide an example of a disaster recovery use case.

3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.
5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Application Based Questions Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity

AUTO - SCALING AND LOAD BALANCING:

Cloud auto-scaling and load balancing help companies handle unpredictable traffic during seasonal sales.

* **Auto-Scaling:**

Automatically increases or decreases the number of cloud servers according to user demand.

* **Load balancing:**

Distributes incoming requests among multiple servers to prevent any single server from becoming overloaded.

* **Reduced downtime:**

If one server fails or becomes overloaded, traffic can be redirected to other healthy servers.

* **Improved user experience:**

Customerms can browse produts and complete payments quickly even during heavy traffic.

8) SECURE CLOUD STORAGE AND DISASTER RECOVERY:

Cloud storage provides secure and reliable storage systems for large volumes of backup data

* Durability:

Cloud providers use reliable storage systems to protect data against loss.

* Data redundancy:

Multiple copies of data are stored across different devices or locations to reduce the risk of data loss.

* Encryption:

Data is encrypted both during transmission and while stored, preventing unauthorised access.

* Access control:

Permissions, authentication and role-based access control ensure that only authorised.

Example:

A hospital stores encrypted patient record backups in cloud storage. If its local server fails, the hospital can restore the records.

ENT DELIVERY NETWORKS & GLOBAL AVAILABILITY:

* A Content Delivery Network consists of geographically distributed servers that deliver content to users quickly.

* It reduces latency by storing frequently accessed content on nearby edge servers.

* It speed the loading and reduces buffering.

* Edge servers help provide a smooth user experience across different regions.

* Cloud providers use multiple regions and availability zones to ensure global availability and reliability.

* If one server fails, another server can provide the content.

4) CLOUD SECURITY, COMPLIANCE AND DATA SOVEREIGNTY:

Financial institutions must protect sensitive customer information and follow applicable regulations when using cloud services.

* Regulatory compliance:

Cloud providers offer security controls and compliance certifications that help organizations meet relevant regulatory requirements.

* Auditing.

Audit reports and security assessments help verify whether systems follow required policies and standards.

* Logging:

Records of user logins, transactions and system changes help detect suspicious activities and support investigations.

* Monitoring:

Continuous monitoring identifies unusual access, Potential attacks and security threats.

5) CONTAINERIZATION, VIRTUAL MACHINES & KUBERNETES.

* Containers vs Virtual Machines:

Containers are lightweight, start quickly and use fewer resources because they share the host operating system kernel.

...e & Kubernetes:

Kubernetes automates container deployment, scaling, and load balancing. It also restarts failed containers and maintains application availability

* Portability:

Containers can run across compatible environments and different cloud platforms with minimal application changes.

* Example:

An online shopping.